

Answer the following questions.

1. Which one of the following statements about static member functions is TRUE?

- a) They can access only static members directly
- b) They require an object instance
- c) They cannot be accessed using class name
- d) They can directly access all non-static members

Answer: a)

2. Why do static member variables in C++ typically require a separate definition outside the class declaration?

- a) Because static members are stored separately from object instance memory and therefore require their own storage allocation
- b) Because static members must be copied into every object created
- c) Because static members cannot be declared inside a class
- d) Because static members are always private by default

Answer: a)

3. What is the output of the following program?

```
#include <iostream>
using namespace std;

class Test {
public:
    static int x;

    Test() {
        x++;
    }
};

int Test::x = 5;

int main() {
    Test t1;
    Test t2;

    cout << t1.x << " ";
    cout << t2.x;

    return 0;
}
```

- a) 5 5
- b) 6 6
- c) 6 7
- d) 7 7

Answer: d)

4. What is the output of the following program?

```
#include <iostream>
using namespace std;

class Counter {
public:
    Counter() {
        count++;
        value++;
    }

    void display() {
        cout << "count = " << count
              << ", value = " << value << endl;
    }

private:
    static int count;
    int value = 0;
};

int Counter::count = 0;

int main() {
    Counter c1;
    Counter c2;

    c1.display();
    c2.display();

    return 0;
}
```

Answer:

count = 2, value = 1

count = 2, value = 1

5. Which access specifier allows access by derived classes while restricting direct external access?
- a) private
 - b) protected
 - c) public
 - d) Both b and c

Answer: b)

6. State True or False. In C++, when a derived class inherits with private inheritance, public members of the base class become private in the derived class.

Answer: True

7. State True or False. In C++, if no access mode is specified while inheriting from a base class using a class declaration, the inheritance mode defaults to protected.

Answer: False

8. What is the correct syntax for a derived class inheriting from a base class in C++?
- a) class Derived extends Base
 - b) class Derived : Base
 - c) class Derived : public Base
 - d) Both b and c are valid

Answer: d)

9. Which of the following represents multilevel inheritance?
- a) Class A is parent of Class B, and Class B is parent of Class C
 - b) Class C inherits from both Class A and Class B
 - c) Class A and Class B both inherit from Class C
 - d) Class A has multiple instances of Class B

Answer: a)

10. What is the output of the following program?

```
#include <iostream>
using namespace std;

class Animal {
public:
    Animal() {
        cout << "Animal ";
    }
};

class Dog : public Animal {
public:
    Dog() {
        cout << "Dog ";
    }
};

int main() {
    Dog d;
    return 0;
}
```

Answer: Animal Dog

11. What is the output of the following program?

```
#include <iostream>
using namespace std;

class Animal {
public:
    Animal(int x) {
        cout << x << " ";
    }
};

class Dog : public Animal {
public:
    Dog() : Animal(10) {
        cout << 20;
    }
};

int main() {
    Dog d;
    return 0; }
```

Answer: 10 20

12. Which one of the following keywords enables runtime polymorphism in C++?

- a) static
- b) override
- c) dynamic
- d) virtual

Answer: d)

13. What is the primary purpose of the `virtual` keyword in C++?

- a) To prevent a function from being overridden in derived classes
- b) To enable late binding (runtime polymorphism)
- c) To make a function run faster
- d) To make a function's memory allocation virtual

Answer: b)

14. State True or False. Virtual functions allow a member function to be called based on the content of the base class pointer rather than its type.

Answer: True

15. What is the output of the following program?

```
#include <iostream>
using namespace std;

class Animal {
public:
    virtual void sound() { cout << "Generic"; }
};

class Dog : public Animal {
public:
    void sound() { cout << "Woof"; }
};

int main() {
    Animal* ptr = new Dog();
    ptr->sound();
}
```

Answer: Woof

16. State True or False. A pure virtual function must have an implementation in the base class.

Answer: False

17. Which of the following demonstrates early binding?

- a) Calling a virtual function through a base pointer
- b) Calling a non-virtual member function
- c) Dynamic dispatch
- d) Runtime polymorphism

Answer: b)

18. State True or False. A base class pointer can point to an object of a derived class.

Answer: True

19. Can a derived class pointer directly point to a base class object?

- a) Yes, always
- b) No, because a base object is not necessarily a derived object
- c) Only with virtual inheritance
- d) Only with multiple inheritance

Answer: b)

20. What is the output of the following program?

```
#include <iostream>
using namespace std;

class Base {
public:
    Base() { display(); }
    virtual void display() { cout << "Base"; }
};

class Derived : public Base {
public:
    Derived() { display(); }
    void display() { cout << "Derived"; }
};

int main() {
    Derived d;
}
```

Answer: BaseDerived

21. Complete the missing part in the following abstract class definition:

```
class Shape {
public:
    _____ draw() = 0;
};

class Circle: public Shape {
public:
    void draw() { cout << "Circle"; }
};
```

Answer:
virtual void